YIFIEYEH ACHESOMIE GONI
AI/MACHINE LEARNING ENGINEER

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SUMMARY

AI & Machine Learning Engineer with hands-on experience designing, developing, and deploying end-to-end AI and Generative AI applications. Passionate about building intelligent AI products that solve real-world problems and eager to contribute to collaborative engineering teams developing next-generation AI solutions...

TECHNICAL SKILLS

Python | Machine Learning | Deep Learning | RAG | NLP | TensorFlow | Keras | Scikit-learn | Pandas | NumPy | XGBoost | LangChain | Hugging Face | FAISS | Streamlit | Git | GitHub | VS Code | Google Colab | Prompt Engineering | Generative AI | Groq API | Open AI API |

PROJECTS

AI CUSTOMER SUPPORT CHATBOT

- Built a Retrieval-Augmented Generation (RAG) chatbot capable of answering questions from uploaded documents by integrating **LangChain, FAISS vector search, Hugging Face embeddings, and Groq LLMs**.
- Integrated **Large Language Models (LLMs)** through API-based inference and implemented prompt engineering techniques to generate accurate, context-aware responses with source citations.
- Reduced response latency by leveraging **Groq's high-speed inference** and efficient vector retrieval for real-time conversational responses.
- Designed a modular AI application using **Streamlit**, with session management, conversation history, source citation, logging, and robust error handling for improved maintainability.

CARDIOVASCULAR DISEASE PREDICTION

- Designed, trained and deployed a deep neural network that predicted cardiovascular disease with **73.2% accuracy and a ROC-AUC of 0.799** by engineering clinical features (BMI, MAP, Pulse Pressure) and optimizing a TensorFlow/Keras model.
- Improved model interpretability by integrating **SHAP explainability**, enabling visualization of feature contributions to individual predictions.

STUDENT PERFORMANCE PREDICTION

- Led a 10-member capstone team to develop and deploy an end-to-end machine learning application for predicting student academic performance.
- The student performance prediction model achieved an **R^2 score of 0.826 and RMSE of 1.52** by comparing Linear Regression and XGBoost models on academic performance data.
- Improved prediction reliability by implementing a robust **scikit-learn preprocessing pipeline** with feature encoding, scaling, and model evaluation.

EDUCATION

Bachelor of Science (B.Sc.) in Biochemistry **March/2024**
Rivers State University, Port Harcourt

CERTIFICATIONS

- **AI/Machine Learning Engineering** – Tech Crush(accredited by **ACTD, USA**)– Unique ID: TEC-3096A0E4
- **Python Certification** – FreeCodeCamp
- **National Youth Service Corps Certificate (NYSC)** – Certificate no: 005984597